Computational Reconsolidation: A Parameterized Mechanism for Memory Drift in Event-Driven LLM Agents

Deliverable A: Executive Summary

The prevailing paradigm in Large Language Model (LLM) agent memory architectures treats stored information as an immutable, append-only ledger or a static knowledge graph. This digital archiving approach contradicts the biological reality of human cognition, where memory is a reconstructive, highly malleable process. This report proposes a mathematically validated, lazy-compatible computational model of memory reconsolidation explicitly designed for the MiroFish/OASIS multi-agent simulation framework.

The core mechanism adapts the Latent Cause Theory of memory modification[1] to LLM embedding spaces. Upon retrieval or social reinforcement, a memory trace enters a transient labile state—subject to modification—only if the current retrieval context presents a moderate Prediction Error (PE) relative to the stored trace[2]. This "sweet spot" of PE acts as a destabilization gate. If destabilized, the memory's semantic gist, emotional valence, and confidence parameters undergo bounded drift toward the new context, governed by a calculated Plasticity factor (P).

This Plasticity factor is inversely proportional to the memory's existing Stability (S) and dynamically modulated by the agent's personality traits (mapped from the Big Five to the TGA model). For instance, high Neuroticism (low TGA Stability) significantly dampens the Fading Affect Bias (FAB), causing negative valences to persist, while high TGA Plasticity (Extraversion/Openness) accelerates semantic drift[4].

To satisfy the simulator's event-driven, lazy-evaluation constraints, background social reinforcements are captured as lightweight embeddings in an exposure_log. These are integrated via a sequential time-decayed update function only when the agent next explicitly retrieves the memory. The original episodic trace remains immutably archived as the ground-truth historical record, while the corresponding "Belief" node undergoes iterative rewriting. This architecture [INFERENCE] provides the first formalized, parameter-driven implementation of the misinformation effect, mood-congruent recall, and therapeutic reappraisal for social simulation agents, effectively operationalizing the "memory is clay" hypothesis. The confidence level in the proposed mathematical model is high for structural validity, though empirical parameter tuning within the simulation remains necessary.

1. The Biological and Psychological Foundations of Reconsolidation (RQ1)

The conceptualization of memory as a permanent, fixed engram was fundamentally disrupted by modern neuroscience, which demonstrated that retrieval returns consolidated memories to

a transiently unstable state. From this state, the memory must restabilize—a process termed reconsolidation[6]. [STRONG EVIDENCE]

1.1 The Discovery of the Labile Window

The existence of a reconsolidation process was systematically proven in canonical auditory fear-conditioning studies. When animals were reminded of a conditioning session by exposure to a conditioned stimulus (CS), the memory trace re-entered a labile state[6]. If a protein synthesis inhibitor (such as anisomycin) was infused into the basolateral amygdala (BLA) immediately following this reactivation, the animals exhibited intact post-reactivation short-term memory (PR-STM) but impaired post-reactivation long-term memory (PR-LTM)[6]. This proved that recalling a memory forces the brain to physically rebuild it.

Crucially, this labile window is strictly time-bounded. Interventions applied six hours post-reactivation have no amnestic effect, demonstrating that the memory has successfully restabilized[3]. Behavioral interventions perfectly parallel pharmacological ones. The Monfils-Schiller paradigm demonstrated that introducing extinction training strictly within the reconsolidation window (e.g., 10 minutes to 1 hour post-retrieval) prevents the spontaneous recovery and reinstatement of fear, essentially overwriting the original fear trace[9]. [STRONG EVIDENCE]

However, the literature surrounding the Monfils-Schiller paradigm contains notable replication failures and intense debate regarding boundary conditions[10]. [MIXED] It has become evident that simply retrieving a memory does not guarantee the opening of the reconsolidation window. The memory must be functionally destabilized, which requires specific environmental triggers.

1.2 Boundary Conditions: Prediction Error, Age, and Strength

Prediction Error (PE)—a mismatch between the organism's prior expectations and the actual events occurring during the reactivation session—is now recognized as the fundamental, necessary boundary condition for memory destabilization[2]. [STRONG EVIDENCE] If the retrieval environment perfectly matches the acquisition environment (zero PE), no updating is required, and the memory remains completely stable. Conversely, if the PE is exceedingly large, the organism infers that it has entered a novel environment and creates a entirely new, parallel memory trace rather than updating the old one[1]. Reconsolidation strictly requires an intermediate degree of PE—a "sweet spot" of mismatch[3].

Furthermore, the intrinsic properties of the memory itself dictate its susceptibility to change. Memory strength and memory age serve as critical boundary conditions[1]. Highly consolidated, older memories, as well as exceptionally strong memories formed with high-intensity emotional or physical stimuli, exhibit profound resistance to destabilization[1]. Overcoming the boundary conditions of an old, strong memory requires either multiple spaced retrieval sessions or a carefully calibrated, higher degree of prediction error[3].

2. Formalizing the Mechanism: Computational Models (RQ2)

Translating the biological phenomena of reconsolidation into computable algorithms requires

robust cognitive modeling. Two primary paradigms provide the mathematical foundation for the proposed LLM agent architecture: Latent Cause Theory and the ACT-R cognitive architecture.

2.1 Latent Cause Models of Memory Modification

Gershman's Latent Cause Theory provides the most comprehensive mathematical framework for reconsolidation, modeling memory modification as an elegant process of Bayesian structural inference[1]. [STRONG EVIDENCE] According to this framework, an organism does not simply record sensory data; it attempts to infer the hidden "latent causes" generating that data[1]. When an agent encounters a stimulus, it calculates the posterior probability of which latent cause generated it using Bayes' rule: $P(cause|data) \propto P(data|cause)P(cause)$ [15]. The model operates on two foundational principles:

1. **The Simplicity Principle:** Sensory inputs tend to be generated by a small (but unbounded) number of latent causes. This is guided by a Bayesian non-parametric infinite-capacity prior, which prefers fewer causes but allows the creation of new ones when absolutely necessary[1].
2. **The Contiguity Principle:** Events occurring closer in time are more likely to have been generated by the same latent cause. Latent causes possess temporal persistence[1].

In the context of reconsolidation and extinction, the Latent Cause Theory mathematically formalizes the "Prediction Error sweet spot." If a retrieval cue yields a small-to-moderate PE, the Bayesian posterior probability strongly favors assigning the new observation to the existing latent cause. The system then updates the associative weights of that cause via a Temporal Difference (TD) delta rule: $\Delta w = \alpha(r_t - \hat{Q})$, where $\hat{Q}$ is the predicted outcome[1]. If the PE is too large, the posterior probability of the existing cause drops, and the prior favors the instantiation of a new latent cause. Consequently, the original memory remains untouched, and a new trace is formed[1].

When mapping this to LLM agents, a "latent cause" corresponds directly to a specific semantic "gist" node in the knowledge graph. The Prediction Error can be quantitatively approximated as the cosine distance in the embedding space between the stored gist and the current prompt context. [INFERENCE]

2.2 Activation and Decay in ACT-R

While Latent Cause Theory handles the structural routing of updates, the temporal dynamics of memory availability are best modeled by the Adaptive Control of Thought-Rational (ACT-R) cognitive architecture[19]. [STRONG EVIDENCE] ACT-R models declarative memory using distinct symbolic chunks, each possessing a continuous, sub-symbolic activation value[19].

A core tenet of ACT-R is that the base-level activation of a memory chunk decays as a power function of time since its last retrieval[21]. Retrieving a chunk introduces noise and requires the chunk's activation to exceed a specific retrieval threshold[23]. The existing MiroFish architecture relies on the Free Spaced Repetition Scheduler (FSRS) equation for retrievability: $R = (1 + t/(9S))^{-1}$. This formulation perfectly aligns with the ACT-R power-law decay dynamic, ensuring that memory degradation scales logarithmically. Integrating reconsolidation

with this equation requires ensuring that retrieval-induced plasticity is scaled by current retrievability (R) and inversely bounded by stability (S).

3. Quantifying Memory Distortion and Drift (RQ3)

Memory reconsolidation is the biological mechanism underlying real-world psychological memory distortions. Designing realistic cognitive agents requires parameterizing these distortion effects based on empirical human data.

3.1 The Misinformation Effect and Multiple Exposures

The profound malleability of human memory is most robustly demonstrated by the misinformation effect, extensively documented by Elizabeth Loftus[25]. [STRONG EVIDENCE] When individuals are exposed to misleading post-event information, this new data actively interferes with and frequently overwrites aspects of the original episodic memory trace[25]. Meta-analytic reviews and registered replications demonstrate substantial effect sizes for misinformation incorporation, with early high-powered replications noting a Cohen's d of 0.75 for the basic misinformation effect[28].

Crucially for social simulations, the effect of *multiple* pieces of misinformation is non-linear. Research into "double misinformation" indicates a complex mechanistic interplay between the "multiple chances" mechanism and the "neutralization" mechanism[29]. When misinformation is subtle and repeated, it increases the probability of memory contamination. However, when an individual encounters blatantly contradictory misinformation, it triggers discrepancy detection (neutralization), which drastically reduces the credibility of the input and protects the original memory[28]. [STRONG EVIDENCE]

This psychological discrepancy detection perfectly mirrors the "Prediction Error sweet spot" finding in neuroscience: if the semantic contradiction (the PE) crosses a maximum threshold, the agent rejects the update.

3.2 Fading Affect Bias and Mood-Congruent Distortions

Memory drift encompasses not just semantic content, but also emotional valence. The Fading Affect Bias (FAB) is a robust psychological phenomenon dictating that the intensity of negative emotions associated with autobiographical memories decays at a significantly faster rate than positive emotions[30]. [STRONG EVIDENCE] This bias serves as an adaptive coping mechanism, promoting long-term emotional resilience.

However, the FAB is not universal; it is highly dependent on cognitive habits. Rumination—the tendency to repetitively and passively focus on symptoms of distress and their causes—severely disrupts the FAB[4]. Individuals who engage in high levels of reflective rumination experience a significantly reduced FAB, resulting in the prolonged persistence of negative memory valence[4]. Meta-analyses show that effect sizes for the relationship between rumination, neuroticism, and negative memory bias are moderate but highly consistent ($r \approx 0.20$ to 0.39)[33].

4. Trait Modulation: Personality and Reappraisal (RQ4)

In the MiroFish simulator, baseline agent personalities are sampled from the Big Five facets

(IPIP-NEO-120) and mapped to the TGA model: Stability (combining Neuroticism, Agreeableness, and Conscientiousness), Plasticity (Extraversion and Openness), and Disinhibition. These overarching traits critically modulate the rate, bounds, and direction of memory drift during the reconsolidation window. [INFERENCE]

4.1 Stability, Neuroticism, and Rumination

The TGA trait of Stability operates inversely to Neuroticism. High Neuroticism is robustly correlated with increased cognitive complaints, rumination, and a severely blunted Fading Affect Bias[4]. [STRONG EVIDENCE] Furthermore, highly neurotic or anxious individuals exhibit a pronounced attentional and recall bias toward threatening, negative, or pain-related stimuli[34]. Consequently, in agents with low TGA Stability, the natural positive drift of negative memories (the FAB) must be mathematically penalized. These agents will tend to lock onto negative valences and integrate negative social reinforcements more rapidly than positive ones, simulating clinical rumination.

4.2 Plasticity, Extraversion, and Cognitive Reappraisal

The TGA trait of Plasticity maps directly to Extraversion and Openness to Experience. These traits are positively associated with cognitive flexibility and the successful deployment of cognitive reappraisal strategies[5]. [STRONG EVIDENCE] Cognitive reappraisal—the active reinterpretation of an emotion-eliciting situation to alter its emotional impact—facilitates rapid schema updating and semantic drift[38].

Agents characterized by high Plasticity will exhibit a larger general learning rate (P) during the labile reconsolidation window. They will more readily incorporate new social reinforcements, accept subtle misinformation without triggering discrepancy detection, and continuously update their memory gists to match their evolving social environments.

5. Existing LLM-Agent Memory Architectures (RQ5)

Traditional LLM memory architectures, such as those utilized in Generative Agents, Mem0, and early versions of Zep, rely entirely on appending new observations to a vector database. During inference, these systems rely on similarity search and LLM attention mechanisms to synthesize the static logs into a coherent prompt context. [STRONG EVIDENCE] This approach fails to model the dynamic degradation and structural merging of memories over time.

Recent advances, notably MemGPT (now Letta), introduce explicit hierarchical memory management. MemGPT allows the LLM agent to output specific function calls (e.g., core_memory_append, core_memory_replace) to edit its own memory blocks, simulating a paging system between working memory and infinite context[40]. [STRONG EVIDENCE]

While MemGPT successfully creates editable state, it treats memory modification as a conscious, executive-function action taken by the LLM. If an agent is lied to, it must consciously reason about the lie and decide to execute a tool call to rewrite its belief. None of the current mainstream architectures natively implement *unconscious, retrieval-induced reconsolidation* governed by prediction error. [INFERENCE] In human cognition, the mere act of retrieving a memory in the context of subtle misinformation unconsciously blends the two, provided the misinformation passes the PE boundary. The architecture proposed below bridges this gap.

6. Deliverable B: Exact Equations for Retrieval-Induced Drift (RQ7)

To operationalize reconsolidation within the LLM agent graph, every retrieval or social reinforcement event opens a potential labile window. The system calculates a Prediction Error, determines if it falls within the destabilization bounds, and applies a calculated, bounded drift to the memory's semantic content, valence, and confidence.

6.1 Variable Definitions and System Context

- S : Memory Stability. Represents resistance to forgetting, calculated via FSRS.
- R : Memory Retrievability ($0 < R \leq 1$). $R = (1 + \Delta t/(9S))^{-1}$.
- V : Memory Valence $[-1.0, 1.0]$, where -1.0 is extreme negative and 1.0 is extreme positive.
- C : Memory Confidence $[0.0, 1.0]$.
- T_{stab} : Agent's TGA Stability trait $[0.0, 1.0]$. (Derived by inverting Neuroticism). Default 0.5 .
- T_{plas} : Agent's TGA Plasticity trait $[0.0, 1.0]$. (Derived from Extraversion/Openness). Default 0.5 .
- $\mathbf{e}_{gist}$: LLM Embedding vector of the current memory's text.
- $\mathbf{e}_{ctx}$: LLM Embedding vector of the new retrieval context or social reinforcement.

6.2 Prediction Error (PE) Calculation

The Prediction Error is quantitatively approximated as the cosine distance between the stored memory gist and the current context. [INFERENCE]

$$PE = 1 - \frac{\mathbf{e}_{gist} \cdot \mathbf{e}_{ctx}}{\|\mathbf{e}_{gist}\| \|\mathbf{e}_{ctx}\|}$$

Note: $PE \in [0, 2]$ *, where* 0 *indicates identical semantics and* 2 *indicates strictly diametric semantics.*

6.3 The Destabilization Gate (Ω)

Destabilization only occurs within the PE "sweet spot"[1]. If $PE < \theta_{min}$, the information is perfectly expected, and no reconsolidation drift occurs (mere retrieval). If $PE > \theta_{max}$, the information is blatantly contradictory, triggering discrepancy detection and neutralizing the drift[28].

$$\Omega(PE) = \begin{cases} 1 & \text{if } \theta_{min} \leq PE \leq \theta_{max} \\ 0 & \text{otherwise} \end{cases}$$

Default Parameters: $\theta_{min} = 0.05$. The upper bound is modulated by personality: $\theta_{max} = 0.35 + (T_{stab} \cdot 0.15)$. Highly stable, agreeable agents are slightly more tolerant of contradiction before rejecting it. [INFERENCE]

6.4 The Plasticity (P) Function

When the destabilization gate is open ($\Omega = 1$), the magnitude of the memory drift is determined by the Plasticity factor P. This factor must be proportional to the agent's trait Plasticity, proportional to the degree of prediction error, and *inversely* proportional to the memory's existing Stability S and Retrievability R (as older, stronger, and more accessible memories resist modification[1]). [INFERENCE]

$$P = \Omega(PE) \cdot (\alpha \cdot T_{plas}) \cdot \left(\frac{PE}{1 + \gamma S} \right) \cdot (1 - R)$$

Default parameters: $\alpha = 0.4$ (Base learning rate scalar, derived from Loftus effect size averages), $\gamma = 0.1$ (Stability resistance scalar). Note the $(1 - R)$ term ensures that highly retrievable, fresh memories resist modification slightly more than fading memories, mimicking the consolidation gradient[7].

6.5 The Update Rules

1. Valence Drift: Valence drift combines the immediate contextual pull with the autonomous Fading Affect Bias (FAB), heavily modulated by TGA Stability[4].

$$V_{new} = V_{old} + P \cdot (V_{ctx} - V_{old}) + \delta_{FAB}$$

Where the fading of negative emotion over time Δt is computed as:

$$\delta_{FAB} = \begin{cases} k \cdot \Delta t \cdot T_{stab} & \text{if } V_{old} < 0 \\ 0 & \text{if } V_{old} \geq 0 \end{cases}$$

(Default $k = 0.005$ *per simulated hour). If an agent has near-zero Stability (high neuroticism),* δ_{FAB} *approaches zero, locking the negative valence in place.*

2. Confidence Drift:

Subtle misinformation decreases confidence in the original specific details, whereas repeated

retrieval in highly congruent contexts (low PE) reinforces confidence. [INFERENCE]

$$C_{new} = \text{clamp}(C_{old} - (P \cdot \omega_{doubt}) + (1 - \text{PE}) \cdot \omega_{reinf},\ 0.1,\ 1.0)$$

(Default $\omega_{doubt} = 0.25$ *,* $\omega_{reinf} = 0.05$ *).*

3. Semantic Gist Drift:
Semantic content drift cannot be achieved via continuous vector arithmetic; it requires discrete LLM generation. To optimize API calls, text generation is only triggered when accumulated P crosses a threshold $P_{threshold}$ (e.g., 0.15).
LLM Prompt Construct: "Rewrite the following memory gist by subtly incorporating elements of the new context. Do not completely overwrite the memory, but blend the new context into the old belief. Degree of change required: {P}."

4. Stability Reinforcement:
Following the reconsolidation process, the memory restabilizes. The act of retrieval and processing structurally increases its long-term stability due to the spacing effect, feeding back into the FSRS system.

$$S_{new} = S_{old} \cdot (1 + e^{W \cdot P})$$

(Where W *is the FSRS difficulty matrix weight for successful recall).*

7. Deliverable C: Lazy-Compatible Formulation

In an event-driven simulator like MiroFish, calculating continuous drift per tick is computationally unviable. Agents spend the vast majority of their time in a dormant state, only activating for LLM inference when contextually relevant. Memory drift must be calculable lazily, preserving mathematical consistency regardless of the length of the time-skip.

7.1 The Exposure Log Architecture

1. **Passive Social Reinforcement:** When Agent B gossips to Agent C about a memory originally owned by dormant Agent A, the simulator does *not* wake Agent A to process reconsolidation.
2. **Appending to the Log:** Instead, the simulator appends a lightweight data dictionary {timestamp, context_text, context_embedding, context_valence} to a list property called exposure_log on Agent A's memory node in the temporal graph.
3. **Calculation at Retrieval:** When Agent A eventually awakens and retrieves the memory (either via similarity search or direct entity traversal), the system executes a pre-retrieval hook.

7.2 Sequential Accumulation

To correctly simulate continuous time, the hook iterates through the exposure_log chronologically. For each entry, it calculates the Retrievability R and δ_{FAB} based strictly on

the Δt between the *previous* log entry's timestamp (or the last_retrieved timestamp) and the current entry's timestamp.

It sequentially updates V, C, S, and accumulates P_{total}. Because S and V are updated at each step, the resistance to modification accurately models the passage of time. [INFERENCE]

7.3 Batch Semantic Update

If the final accumulated P_{total} exceeds the semantic threshold, a single LLM generation call is made. The prompt synthesizes all context_text elements from the log that successfully passed the destabilization gate to rewrite the memory gist. Following this, the exposure_log is cleared, and the node's last_retrieved timestamp is updated to the current time. This guarantees O(1) LLM calls per actual retrieval, regardless of how many background social reinforcements occurred.

8. Deliverable D: Personality-Modulation Table

This table defines how the TGA traits (derived from the Big Five) specifically and mathematically modulate the reconsolidation parameters. [INFERENCE based on psychological literature]

Trait	Target Parameter	Mathematical Effect	Evidence Base
Stability (Low Neuroticism)	δ_{FAB} (Fading Affect Bias)	High Stability linearly increases the rate at which negative valence shifts toward neutral. Low Stability (Rumination) locks negative valence.	STRONG EVIDENCE[4]
Stability (High Agreeableness)	θ_{max} (Destabilization Bound)	High Stability increases tolerance for contradiction ($\theta_{max} = 0.35 + T_{st}$), making the agent more susceptible to social misinformation before detecting a	SPECULATION

		discrepancy.	
Plasticity (Openness/Extraversion)	α (Base Plasticity Multiplier)	High Plasticity increases the magnitude of semantic and valence drift given the same PE, mathematically modeling superior cognitive reappraisal and schema flexibility.	STRONG EVIDENCE[5]
Plasticity	ω_{reinf} (Confidence Gain)	Low Plasticity increases cognitive rigidity; the agent gains confidence faster from congruent retrievals, becoming dogmatic and resistant to future changes.	INFERENCE
Disinhibition	$P_{threshold}$ (LLM Trigger)	High Disinhibition lowers the threshold required to trigger a full semantic rewrite via the LLM, leading to more erratic, rapidly shifting narratives.	SPECULATION

9. Deliverable E: Graph Schema Modifications (Neo4j/Graphiti) (RQ6)

Implementing reconsolidation safely in Neo4j or Graphiti requires maintaining strict data provenance. Without provenance, the system risks runaway hallucination, making it impossible to distinguish between the objective historical record and the agent's subjective, drifted

belief[29].

9.1 Schema Additions

The graph schema must separate immutable events from mutable beliefs:

- **EpisodicEvent (Immutable):** Represents the ground-truth backstory, physical action, or simulator event. This node is created once and never modified.
- **Belief (Mutable):** The semantic gist node that undergoes reconsolidation.
  - *New Properties:* valence (float), confidence (float), stability (float), last_retrieved (int epoch), exposure_log (JSON string/array).
  - *Temporal Fields:* valid_at and invalid_at. Crucially, invalid_at is strictly reserved for when a fact becomes functionally false in the simulator's world state (e.g., "The building burned down"), *not* for psychological forgetting or drift.
- **BeliefVersion (Immutable History):** Whenever a Belief undergoes semantic drift via the LLM, the previous text and parameters are cloned into a BeliefVersion node.

9.2 Provenance Edges

To maintain strict lineage, the following edges must be maintained:

- (Belief)-[:GROUNDED_IN]->(EpisodicEvent)
- (Belief)-[:HAS_PREVIOUS_VERSION]->(BeliefVersion)

This versioned architecture serves as a critical safeguard against memory contamination. If an agent falls into a gossip-amplification feedback loop and hallucinations run out of control, analytical tools (or explicit debugging commands) can traverse the HAS_PREVIOUS_VERSION chain back to the GROUNDED_IN event to reconstruct the original truth. [INFERENCE]

10. Deliverable F: Pseudocode Implementation

The following Python pseudocode demonstrates the lazy evaluation of memory drift upon retrieval. *(Note: Code logic is designed for mathematical and logical validity; exact integration into the MiroFish SDK requires structural adaptation. Untested in a production environment).*

10.1 Python Logic

```
Python
import numpy as np
from dataclasses import dataclass
from typing import List

@dataclass
class ExposureEvent:
    timestamp: int
    context_text: str
    context_embedding: np.ndarray
    context_valence: float
```

```
class BeliefNode:
    def __init__(self, node_id, text, gist_embedding, valence, confidence, stability, last_retrieved, owner_stability, owner_plasticity):
        self.node_id = node_id
        self.text = text
        self.gist_embedding = gist_embedding
        self.valence = valence
        self.confidence = confidence
        self.stability = stability
        self.last_retrieved = last_retrieved
        self.exposure_log: List[ExposureEvent] = []

        # TGA Personality traits
        self.t_stab = owner_stability
        self.t_plas = owner_plasticity

    def cosine_distance(self, emb1, emb2):
        """Calculates Prediction Error (PE)"""
        return 1.0 - (np.dot(emb1, emb2) / (np.linalg.norm(emb1) * np.linalg.norm(emb2)))

    def process_lazy_reconsolidation(self, current_time: int, llm_client):
        """Pre-retrieval hook to process accumulated drift sequentially."""
        if not self.exposure_log:
            # Update FAB and Retrievability up to current_time even if no exposures
            delta_t_final = current_time - self.last_retrieved
            self._apply_fab(delta_t_final)
            self.last_retrieved = current_time
            return

        cumulative_plasticity = 0.0
        contexts_for_rewrite = []

        # Sort chronologically to simulate continuous passage of time
        self.exposure_log.sort(key=lambda x: x.timestamp)

        for event in self.exposure_log:
            delta_t = event.timestamp - self.last_retrieved

            # 1. Apply natural fading prior to the exposure event
            self._apply_fab(delta_t)

            # 2. Calculate Retrievability (FSRS power law)
```

```
            R = 1.0 / (1.0 + (delta_t / (9.0 * self.stability)))

            # 3. Calculate Prediction Error
            pe = self.cosine_distance(self.gist_embedding, event.context_embedding)

            # 4. Destabilization Gate (The PE Sweet Spot)
            theta_max = 0.35 + (self.t_stab * 0.15)
            if 0.05 <= pe <= theta_max:
                # 5. Calculate Plasticity
                alpha = 0.4
                gamma = 0.1
                P = (alpha * self.t_plas) * (pe / (1.0 + gamma * self.stability)) * (1.0 - R)
                cumulative_plasticity += P

                # 6. Valence Drift
                self.valence = self.valence + P * (event.context_valence - self.valence)
                self.valence = max(-1.0, min(1.0, self.valence))

                # 7. Confidence Drift
                self.confidence = max(0.1, min(1.0, self.confidence - (P * 0.25)))

                # 8. Restabilization (increases resistance to future edits)
                self.stability = self.stability * (1.0 + np.exp(0.5 * P))

                contexts_for_rewrite.append(event.context_text)

            # Congruent reinforcement (PE < 0.05) increases confidence without rewriting
            elif pe < 0.05:
                self.confidence = min(1.0, self.confidence + 0.05)

            self.last_retrieved = event.timestamp

        # 9. Semantic Drift (Triggered if accumulated P is high enough)
        if cumulative_plasticity > 0.15 and contexts_for_rewrite:
            combined_context = " ".join(contexts_for_rewrite[-3:]) # Use recent contexts
            self.trigger_semantic_rewrite(combined_context, cumulative_plasticity, llm_client)

        # Clean up log and sync time
        self.exposure_log = []
        delta_t_final = current_time - self.last_retrieved
        self._apply_fap(delta_t_final)
        self.last_retrieved = current_time
```

```
    def _apply_fab(self, delta_t):
        """Applies the Fading Affect Bias to negative memories."""
        if self.valence < 0:
            # delta_t assumed in hours for k scalar
            delta_fab = 0.005 * delta_t * self.t_stab
            self.valence = min(0.0, self.valence + delta_fab)

    def trigger_semantic_rewrite(self, context, P, llm_client):
        """Abstract method for invoking LLM and handling Graph versioning."""
        prompt = f"Rewrite the memory '{self.text}' by subtly incorporating the new context: '{context}'. Do not replace the memory entirely, but blend them. Degree of change required (0-1): {min(P, 1.0)}."
        # Call LLM, update self.text, recalculate self.gist_embedding.
        pass
```

10.2 Corresponding Cypher (Neo4j) Structure Updates

To append a passive social reinforcement exposure into a dormant agent's memory node, the simulator executes the following Cypher query:

```
Cypher
MATCH (b:Belief {id: $belief_id})
SET b.exposure_log = coalesce(b.exposure_log, []) +
  '{ "timestamp": ' + $time + ', "context_text": "' + $text + '", "context_embedding": ' + $emb +
  ', "context_valence": ' + $val + ' }'
```

When semantic rewrite occurs, the versioning is executed via:

```
Cypher
MATCH (b:Belief {id: $belief_id})
CREATE (bv:BeliefVersion {
  text: b.text,
  valence: b.valence,
  archived_at: $time
})
CREATE (b)-[:HAS_PREVIOUS_VERSION]->(bv)
// (Followed by updating b with the new LLM output)
```

11. Validation Plan and Failure Modes (Deliverable G & RQ6)

Implementing continuous memory reconsolidation in multi-agent environments introduces complex system dynamics. Validation requires isolated sandbox testing against standard baseline behaviors to ensure mathematical stability.

11.1 Validation Experiments

1. The Misinformation Paradigm Replication:

- **Setup:** Seed Agent A with a clear episodic memory ("Saw a red car at the intersection"). Expose Agent A to Agent B (who holds the conflicting belief: "Saw a blue car") through N social reinforcement interactions over simulated days.
- **Measurement:** Track the cosine similarity of Agent A's memory gist to the target phrase "blue car" over N retrievals.
- **Acceptance Threshold:** The semantic drift should follow a logarithmic curve bounded by confidence limits, rather than a linear or single-shot overwrite. Agent A should eventually report the car as "bluish" or "dark", but must not wholly adopt "blue" unless $N > 5$ and Agent A possesses high Plasticity.

2. Extinction vs. Reconsolidation Ablation:

- **Setup:** Compare the proposed parameterized drift model against a baseline "append contradictory memory" model (as seen in standard Generative Agents or Mem0). Expose both to a rapid change in social standing of a third agent.
- **Measurement:** Frequency of logical contradictions in generated dialogue.
- **Acceptance Threshold:** The baseline append-only model will frequently hallucinate by retrieving the old status and new status simultaneously. The proposed reconsolidation model must reduce the occurrence of logical contradictions in dialogue by >40% by actively resolving and blending discrepancies during the drift phase.

11.2 Mitigating Failure Modes

1. **Runaway Drift and Gossip Amplification:** If agents continuously gossip in a closed loop, memories could rapidly drift into chaotic hallucination (echo chambers).
   - *Safeguard:* The θ_{max} destabilization bound strictly prevents drift from wildly inaccurate inputs. Additionally, the restabilization math (S_{new}) ensures that every time a memory is modified, its Stability increases. This makes subsequent modifications exponentially harder, biologically mimicking how heavily retrieved memories eventually crystalize and become immune to further reconsolidation[6].
2. **Personality Collapse:** Unbounded valence drift could flatten the emotional landscape, pushing all agents toward neutral valences over time.
   - *Safeguard:* The delta_fab calculation is capped by Trait Stability. Furthermore, ensuring that highly positive memories do not decay toward neutral (as FAB primarily affects negative valences) preserves joy and affinity in the graph[30].

12. Open Problems and Unverified Claims (Deliverable H)

While the proposed mechanism is structurally sound and rooted in established cognitive math, several claims regarding the translation of human architectures to LLM embeddings remain empirically unverified within the scope of current computer science literature.

- **Prediction Error to Cosine Similarity Mapping [UNVERIFIED]:** The foundational assumption that Bayesian Prediction Error maps linearly or monotonically to cosine distance in LLM embedding spaces (e.g., OpenAI text-embedding-3-small) is a speculative computational proxy. Embeddings capture semantic similarity, which may not perfectly represent the *predictive* mismatch required to open biological lability windows[2]. Semantic distance may fail to capture structural logic errors.
- **Quantifying the Destabilization Gate [OPEN PROBLEM]:** Finding the exact numerical bounds for the PE "sweet spot" (θ_{min} and θ_{max}) that organically mimics human reconsolidation will require extensive, iterative hyperparameter sweeping during simulation. The default values provided are theoretical starting points.
- **LLM "Subtle" Rewriting Limitations [OPEN PROBLEM]:** Instructing an LLM to "subtly" blend a memory by a calculated float factor of P relies heavily on the advanced instruction-following capability of the local open-weight model (e.g., Qwen). LLMs frequently struggle with continuous, granular semantic shifts, tending toward discrete, binary updates (either perfectly copying the old text or completely rewriting it). Output steering via LoRA or activation patching during the rewrite step may be necessary to enforce genuine gradualism.
- **Cross-Modality of Latent Causes [UNVERIFIED]:** Gershman's models[1] note that spatial proximity and sensory modality heavily affect latent cause inference. In a text-based LLM simulation, spatial, visual, and temporal contexts are flattened into text tokens, potentially reducing the fidelity and accuracy of the structural learning algorithm compared to biological models.

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